

1 Result and discussion

In this summary, we present the outcomes of our experiment employing various base and small transformer models. While our primary focus revolves around tasks related to detecting hate speech in this study, we also evaluated these models deployability. In essence, our findings reveal surprising performance of small transformer models that are very close to base models.

1.1 Evaluation of Transformer-based Models

In this study, the performance of three state-of-the-art transformer architectures—mBERT (bert-base-multilingual-uncased), Bangla-BERT (bangla-bert-base), and XLM-RoBERTa (xlm-roberta-base)—was systematically evaluated for hate speech detection. Unlike traditional sequential models like BiLSTM or BiGRU, these transformer models leverage self-attention mechanisms to capture deep bidirectional contextual dependencies across multilingual and mono-lingual Bengali corpora.

The experiments were conducted using a five-fold cross-validation strategy to ensure the robustness of the findings. Models were fine-tuned by varying critical hyperparameters, including batch sizes (16, 32, 64), learning rates ($1e-5$, $2e-5$, $3e-5$), and training durations (up to 30 epochs). Performance was primarily assessed using Accuracy and Macro F1-score ($M.F1$).

As summarized in Table 1, the XLM-RoBERTa model consistently outperformed other architectures. The best performance was achieved with a batch size of 64 and a learning rate of $3e-5$, reaching a peak accuracy of 85.34% and a Macro F1-score of 0.8533. This suggests that the large-scale cross-lingual pre-training of XLM-RoBERTa provides superior feature representations for the nuances of Bengali hate speech compared to mBERT and the specialized Bangla-BERT. The comprehensive evaluation results for all hyperparameter configurations across the three models are consolidated in Table 2.

Table 1: Comparative Performance of Best-Performing Transformer Configurations.

Model	Params	Batch Size	Learning Rate	Accuracy	Macro F1
XLM-RoBERTa (Base)	270M	64	$3e-5$	0.8534	0.8533
mBERT (Cased)	180M	64	$3e-5$	0.8440	0.8439
Bangla-BERT	110M	64	$3e-5$	0.8437	0.8437

1.2 Evaluation of Small Models

A series of experiments were conducted to evaluate lightweight or distilled pre-trained transformer models, which are optimized for computational efficiency while maintaining high performance. This study focused on Mixed-Distil-BERT, Xtremedistil-L12-H384, and sahajBERT. Each model was evaluated using a five-fold cross-validation strategy, with hyperparameter tuning focusing on batch sizes (16, 32, 64), learning rates ($1e-5$, $2e-5$, $3e-5$), and varying epoch counts (10, 20, 30).

The evaluation metrics primarily focused on validation Accuracy and Macro F1-score ($M.F1$). Among the tested small models, sahajBERT demonstrated the strongest overall performance, achieving a peak accuracy of 0.8507 and a Macro F1-score of 0.8507 when configured with a batch size of 32 and a learning rate of $3e-5$. This suggests that sahajBERT, despite its compact nature, is highly effective at capturing the linguistic characteristics of the dataset. Mixed-Distil-BERT followed closely, reaching a maximum accuracy of 0.8470 with a smaller batch size of 16.

In contrast, the Xtremedistil-L12 model showed comparatively lower performance, with its best configuration (Batch 16, LR $3e-5$) reaching an accuracy of 0.8245. This discrepancy indicates that while distilled models offer efficiency, the architecture of sahajBERT and the mixed distillation approach of Mixed-Distil-BERT are better suited for this specific classification task. The maximum

Table 2: Consolidated Hyperparameter Search Results Grouped by Model.

Model Architecture	Batch	LR	Epochs	Val Acc	Val M.F1
XLM-RoBERTa (Base)	64	$3e-5$	30	0.8534	0.8533
	32	$3e-5$	30	0.8516	0.8516
	16	$3e-5$	30	0.8525	0.8524
	64	$2e-5$	20	0.8513	0.8511
	32	$2e-5$	20	0.8263	0.8261
	16	$2e-5$	20	0.8288	0.8287
	64	$1e-5$	10	0.8492	0.8492
	32	$1e-5$	10	0.8077	0.8077
	16	$1e-5$	10	0.8187	0.8187
mBERT (Cased)	64	$3e-5$	30	0.8440	0.8439
	32	$3e-5$	30	0.8397	0.8397
	16	$3e-5$	30	0.8394	0.8393
	64	$2e-5$	20	0.8437	0.8437
	32	$2e-5$	20	0.8739	0.8388
	16	$2e-5$	20	0.8437	0.8435
	64	$1e-5$	10	0.8373	0.8369
	32	$1e-5$	10	0.8391	0.8391
	16	$1e-5$	10	0.8415	0.8414
Bangla-BERT	64	$3e-5$	30	0.8437	0.8437
	32	$3e-5$	30	0.8428	0.8428
	16	$3e-5$	30	0.8400	0.8400
	64	$2e-5$	20	0.8419	0.8418
	32	$2e-5$	20	0.8419	0.8418
	16	$2e-5$	20	0.8413	0.8413
	64	$1e-5$	10	0.8361	0.8361
	32	$1e-5$	10	0.8400	0.8398
	16	$1e-5$	10	0.8394	0.8393

performance achieved by each model is summarized in Table 3, and the full hyperparameter search results are consolidated in Table 4.

Table 3: Best-Performing Configuration and Model Size for each Small Model Architecture.

Model Architecture	Batch Size	Learning Rate	Acc	M.F1	Params
sahajBERT	32	$3e^{-5}$	0.8507	0.8507	~18M
Mixed-Distil-BERT	16	$1e^{-5}$	0.8470	0.8470	~66M
XtremeDistil-L12	16	$3e^{-5}$	0.8245	0.8244	~33M

Table 4: Consolidated Hyperparameter Search Results for Small Models.

Model Architecture	Batch	LR	Epochs	Val Acc	Val M.F1
Mixed-Distil-BERT	16	$1e - 5$	10	0.8470	0.8470
	64	$2e - 5$	20	0.8461	0.8460
	32	$3e - 5$	30	0.8370	0.8367
	32	$2e - 5$	20	0.8443	0.8443
	64	$3e - 5$	30	0.8422	0.8421
	16	$2e - 5$	20	0.8428	0.8428
	16	$3e - 5$	30	0.8397	0.8396
	32	$1e - 5$	10	0.8431	0.8431
	64	$1e - 5$	10	0.8403	0.8403
XtremeDistil-L12	16	$3e - 5$	30	0.8245	0.8244
	32	$3e - 5$	30	0.8208	0.8207
	16	$1e - 5$	10	0.8016	0.8016
	64	$3e - 5$	30	0.8029	0.8026
	16	$2e - 5$	20	0.8010	0.8008
	32	$1e - 5$	10	0.7921	0.7921
	32	$2e - 5$	20	0.7845	0.7845
	64	$2e - 5$	20	0.7754	0.7754
	64	$1e - 5$	10	0.7333	0.7329
sahajBERT	32	$3e - 5$	30	0.8507	0.8507
	16	$3e - 5$	30	0.8501	0.8501
	16	$1e - 5$	10	0.8458	0.8457
	32	$2e - 5$	20	0.8428	0.8428
	16	$2e - 5$	20	0.8428	0.8427
	32	$1e - 5$	10	0.8410	0.8409